Academic Project 1 Monitoring

Zero Hunger UM – YUM Hub

A Food Sharing and Monitoring Platfrom for Universiti Malaya Community

Supervisor: Dr. Chiam Yin Kia

Team Members: Queh Qian Yu (22112617)

Subtitle 1: A food-sharing and food-receiving mobile app for the university community

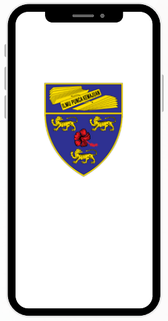
Lim Qian Hui (22091617)

Subtitle 2: A web-based administrator platform for food sharing monitoring and data analysis

Mission Overview

The Zero Hunger platform is a comprehensive food-sharing ecosystem designed to connect surplus food providers with campus members in need.

Through real-time mobile and web technology, we aim to bridge the gap between event over-provisioning and student food insecurity without stigma or restrictions.



Mobile App: Real-time sharing & claiming

Web App: Data analytics & monitoring

Project Background

Target: Providers

University program organizers, staffs and workshop hosts who frequently experience surplus after events.

Target: Receivers

UM Students and staff facing financial constraints or food insecurity due to rising living costs on campus.

Source: United Nations Sustainable Development Goals – Zero Hunger

Problem Statement

Informal Communication

Current efforts rely on fragmented WhatsApp or Telegram groups. These lack a proper channel, leading to limited reach and missed opportunities for redistribution.

(Interview with Pn. Mai, initiator of Zero Hunger UM initiative, 2026)

Rising Living Costs

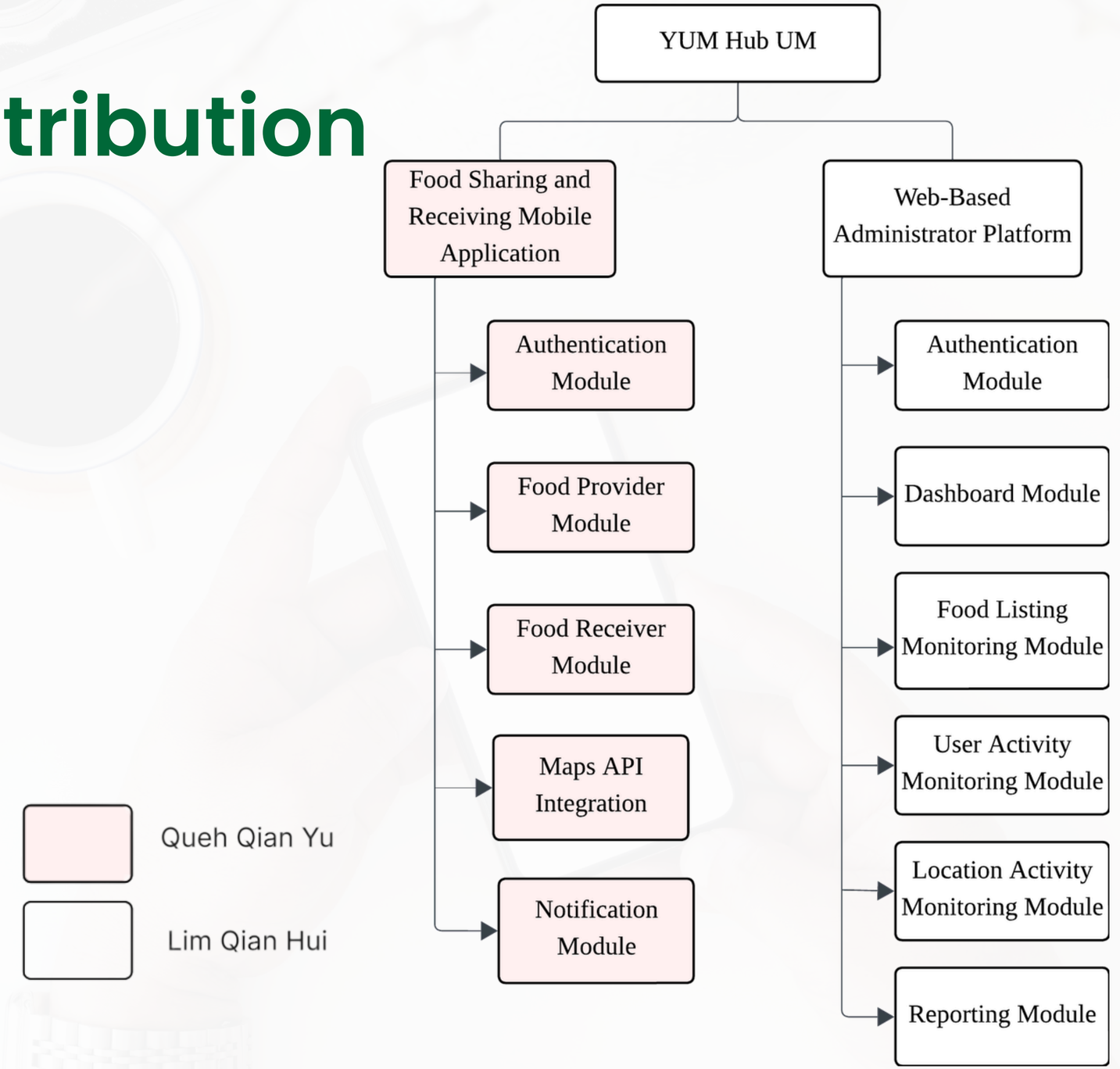
Economic pressure forces students to skip meals or prioritize low-quality nutrition, negatively impacting health and academic performance.

(ChePa et al., 2025)

Project Objective

- To identify the key gaps in connecting food providers and receivers within a campus ecosystem and the factors contributing to food waste.
- To develop an integrated platform that enables seamless coordination between food providers and recipients for efficient food redistribution.
- To test the effectiveness of the proposed system in reducing campus food waste while ensuring safe, timely and accessible food sharing.

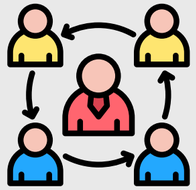
Job Distribution



Queh Qian Yu

Lim Qian Hui

Problem Statement (Subtitle 2)



Inefficient Coordination in Food Aid Management

Fragmented communication between departments creates bureaucratic delays and consumes significant administrative time.

Derivation Source: Literature Review (Reference: Smart Food Donation and Surplus Redistribution System (Buchande et al., 2024))

Lack of Real-Time Monitoring

No centralized platform for real-time tracking of food availability and distribution status results in delayed decision-making and outdated information.

Derivation Source: Literature Review (Reference: Tracking SDG 2 - Zero Hunger - ArcGIS StoryMaps and The State of Food Security and Nutrition in the World 2022 (FAO, 2022))

Problem Statement (Subtitle 2)

Limited Data Analytics Capability

Fragmented communication between departments creates bureaucratic delays and consumes significant administrative time.

Derivation Source: Literature Review (Reference: The Role of Analytics in Achieving the Sustainable Development Goal of Zero Hunger (Peters et al., 2024))

Inequitable Access to Food Support

Exisiting food bank systems are not uniformly accessible by all community members, leads to uneven coverage of welfare assistance.

Derivation Source: Literature Review (Reference: Tracking SDG 2 - Zero Hunger - ArcGIS StoryMaps and The State of Food Security and Nutrition in the World 2022 (FAO, 2022))

Project Objective (Subtitle 2)

- To identify effective data visualization and reporting techniques for monitoring food distribution systems through literature review.
- To design and develop a real-time web-based analytics dashboard for food distribution and user activity monitoring.
- To test the usability, functionality, and security of the system in supporting evidence-based decision-making and equitable food access for the campus community.

Literature Review (Subtopic 2)

Title: Smart Food Donation and Surplus Redistribution System

Authors: Mr. Aditya Suresh Buchande, Mr. Mohd Ajaj Ali Ahmed Choudhari, Mr. Manish Ravindra Bairagi, Mr. Swapnil Dadaso Yadav

Year: 2024

Research Objective	Methodology	Key Findings	Relevance to Study	Strengths	Limitations
To streamline coordination in food aid management by digitizing workflows between donors and agencies to reduce bureaucratic delays	Three-tier architecture using Bootstrap, PHP 8.1, and MySQL; implementation of Role-Based Access Control (RBAC) to manage departmental permissions	Transitioning from informal coordination (phone/social media) to a structured digital platform reduces response times by 30–38%	Justifies the transition from manual, fragmented communication to a centralized multi-role web interface	Database-level UNIQUE constraints prevent race conditions, such as two departments claiming the same food listing	Current AJAX polling architectures introduce a 30-second latency in notification delivery

Literature Review (Subtopic 2)

Title: Foodernity: A Mobile and Web Application for Food Sharing

Authors: John Amiel R. Morilla, Fhillip Carl Bagsic, Mark Kenneth Dela Cruz, Carl Daniel A. Patio, Emeliza R. Yabut

Year: 2021

Research Objective	Methodology	Key Findings	Relevance to Study	Strengths	Limitations
To develop a centralized platform for real-time monitoring of food availability and distribution status.	Agile Model development (6 phases) including system design and functional testing; evaluated via the System Usability Scale (SUS)	Successfully created an admin web module for monitoring all food donation records and inventory statistics	Provides a blueprint for a university-born monitoring system specifically for donor-beneficiary connectivity	High usability scores (3.05 mean) indicate the platform is functional and working properly for users	Monitoring is limited if donors fail to provide manual status updates or input inaccurate food data

Literature Review (Subtopic 2)

Title: Global Hunger and Food Waste Tracking Platform: A Technical Implementation Study
Authors: Rohit Sharma
Year: 2024

Research Objective	Methodology	Key Findings	Relevance to Study	Strengths	Limitations
To integrate data analytics and visualization tools to support evidence-based decision-making in food distribution.	Technical analysis of a multi-cloud system (AWS, GCP, Azure) processing 2.5 petabytes of supply chain data annually	Dashboards achieved a 22% reduction in food waste and 89% accuracy in shortage prediction via ensemble learning	Proves that React-based visual analytics transform raw data into actionable insights for resource allocation	Significant economic benefits ($432M) generated through predictive allocation and optimization	High-scale systems require massive computational resources and complex data ingestion layers

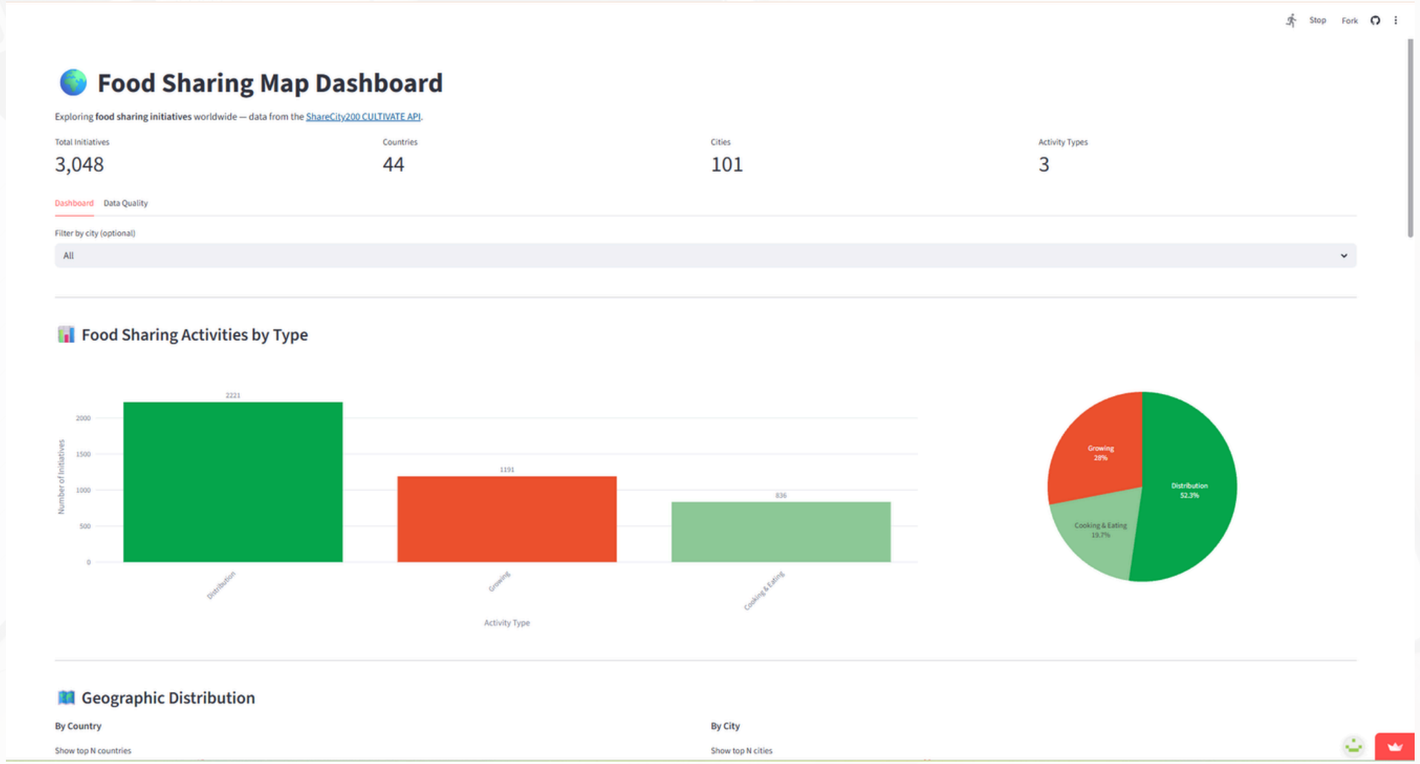
Literature Review (Subtopic 2)

Title: Tracking SDG 2 - Zero Hunger

Research Objective	Methodology	Key Findings	Relevance to Study	Strengths	Limitations
To ensure equitable access to food support by using geo-spatial tools to map and bridge welfare coverage gaps.	Integration of near real-time monitoring via mobile technology (mVAM) with AI-based predictive models	Signals deteriorating situations across indicators like food consumption and market access in over 90 countries	Addresses "uneven coverage" by allowing admins to visualize hotspots and identify underserved locations in real-time	Continuous daily data collection allows for immediate programmatic response to shifting hunger crises	Effectiveness depends on the availability of historical data points (e.g., rainfall, conflict) for AI training

Existing System Comparison

Food Sharing Map Dashboard



Food Sharing Map Dashboard

- Food Sharing Activities by Type
- Geographic Distribution
- Data visualization dashboard
- Food sharing methods
- Pipeline refresh

Existing System Comparison

Proposed YUM Hub UM Dashboard

Improvements

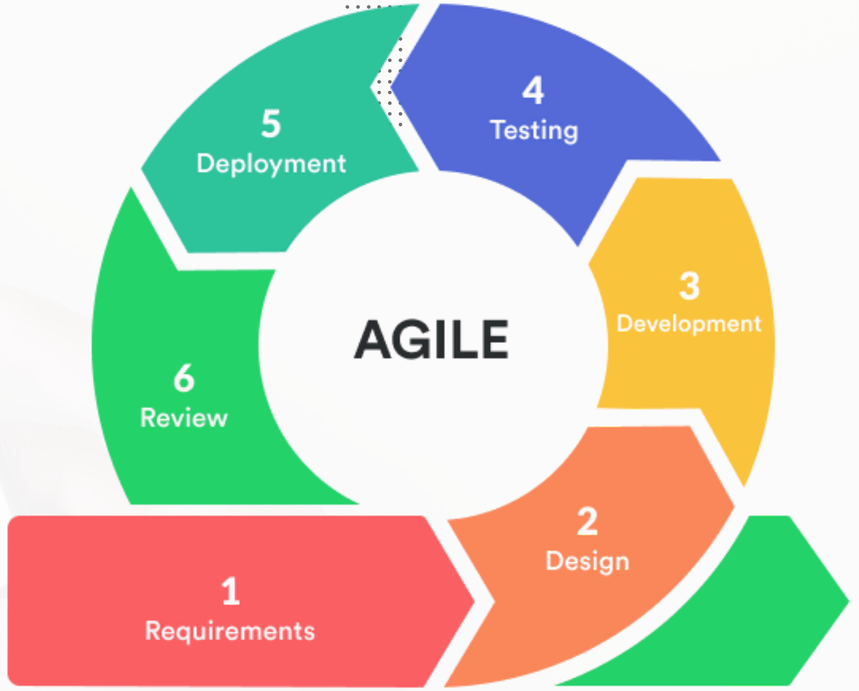
- Adds authentication system
- Introduces admin-level control and management features
  - Manage food listings
  - Monitor user activity and rankings
  - Restrict suspicious users
- Provides real-time operational monitoring
- Enhances dashboard with KPIs
- Includes report generation module
- Shifts system focus from data visualization → decision-making & system governance

Methodology

Agile Scrum Development Methodology

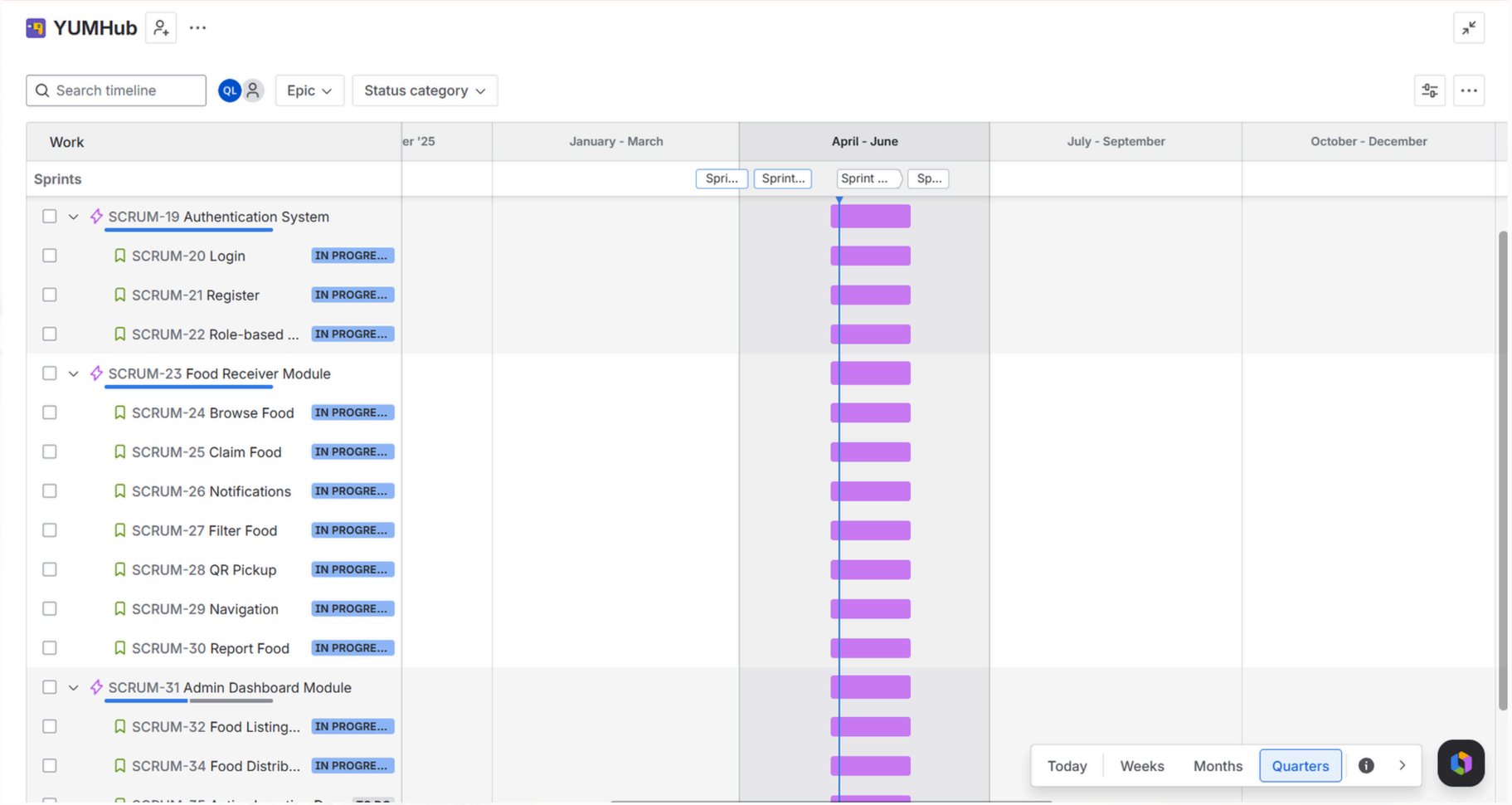
Software Process Model: Agile - Scrum
Sprint Duration: 2-4 weeks per sprint

- Iterative and incremental development approach
- System developed in small cycles called sprints
- Continuous user feedback and improvement
- Flexible to requirement changes



Methodology

Agile Scrum Development Methodology



Stakeholder Collaboration

Jabatan Hal Ehwal Pelajar (HEP), Universiti Malaya

Introduction

- Main student affairs division managing student welfare, development and campus life.
- Supports balanced student life through academic and non-academic development.
- Acts as a bridge between students and university administration for services, activities and welfare.

ROLE AS STAKEHOLDER

- Provide insights on student needs and food insecurity issues
- Validate the relevance of the system within UM campus environment
- Support alignment with existing student welfare initiatives
- Assist in evaluating feasibility and real-world applicability

Benefits of Stakeholder Collaboration

- Ensure the system **aligns with real campus welfare** operations and student needs.
- Provides **expert validation** for system practicality and real deployment in the university.
- Supports **reduction of food waste and improves food accessibility** within the campus community.

Literature Review

Stakeholder Interview

Complete Interview Questions

Project Implementation: The Food Claim Workflow

Our implementation focuses on a real-time synchronization loop between the Food Provider and the Food Receiver. When a Provider uploads surplus food, the system generates a listing with information of the food (types/pax/location to pick-up food). The core logic follows a "First-Claim, First-Served" basis: the moment a student clicks "Claim," the system decrements the available count and locks that portion to the student's ID. This prevents over-claiming and ensures that the "Sold Out" status is triggered instantly when the counter reaches zero. To close the loop, we implement a QR-based validation at the pickup site, which marks the transaction as "Completed" in the Admin dashboard for data tracking.

QUESTIONS:
1. Understanding Current Processes & Pain Points

1. How is surplus food currently managed after a major university event? Is there an existing informal channel (like WhatsApp or Telegram) used to blast notifications?"

2. What is the protocol if food is *still* left over after an informal announcement (e.g., in a WhatsApp group)? Who is responsible for the disposal, and how much of a burden is this cleaning process for the staff?

3. Do you find that current methods (Telegram/WhatsApp) lead to 'information overload' or confusion, such as students arriving for food that was finished 30 minutes ago?

Stakeholder Collaboration Initiative/Letter

UNIVERSITI MALAYA

17th April 2026

Jabatan Hal Ehwal Pelajar (*Student Affairs Department*)
Aras 2, Blok E, Kompleks Perdanasiswa
Universiti Malaya
50603 Kuala Lumpur.

Sir/ Madam,

Collaborator for Academic Project Courses, Faculty of Computer Science and Information Technology, Universiti Malaya

With reference to the above matter,

2. The Faculty of Computer Science and Information Technology (FCSIT) University Malaya is pleased to invite you to be the collaborator for the **Zero Hunger UM – Food Sharing App for Universiti Malaya Community.** This project is under the supervision of **Dr. Chiam Yin Kia.**

3. The Academic Project courses are mandatory for final year students of the Faculty of Computer Science and Information Technology as part of their graduation requirements. These courses are designed to nurture academically proficient and technically skilled graduates in the fields of Computer Science and Information Technology. During the course, students actively apply technical expertise, critical thinking, and problem-solving skills to develop innovative products. These projects are intended to meet collaborators' needs and enhance organizational efficiency.

4. This collaboration entails the involvement of the following student(s) from the Faculty of Computer Science and Information Technology working alongside you on the implementation of the aforementioned project. The faculty kindly requests your support in providing the student(s) with necessary information and/or other related resources to facilitate the successful completion of the project.

5. The following are the student(s) who will be involved in this collaboration

Name	**: Queh Qian Yu**
Matric Number	**: 22112617**
Email	: 22112617@siswa.um.edu.my
Name	**: Lim Qian Hui**
Matric Number	**: 22091617**
Email	: 22091617@siswa.um.edu.my

Faculty of Computer Science and Information Technology
Universiti Malaya, 50603 Kuala Lumpur, MALAYSIA
Tel: (+603) 7967 6300/6301 • http://fsktm.um.edu.my

6. We sincerely hope you will give thoughtful consideration to this invitation. We are confident that this collaboration will bring mutual benefits to both the faculty and your organization. Additionally, we kindly request your written consent to formalize your participation in this initiative. Should you require any further information, please do not hesitate to contact the project supervisor, **Dr. Chiam Yin Kia**, at yinkia@um.edu.my.

Thank you.

Yours sincerely,

Professor Dr. Nor Liyana Mohd Shuib
Deputy Dean (Undergraduate)
Faculty of Computer Science and Information Technology
Universiti Malaya
Kuala Lumpur

Dr. Chiam Yin Kia
Project Supervisor
Faculty of Computer Science and Information Technology
Universiti Malaya
Kuala Lumpur

Acceptance of Collaboration

I hereby acknowledge that I accept this proposal for collaboration with the Faculty of Computer Science & Information Technology, Universiti Malaya pertaining to the research project entitled **Zero Hunger UM – Food Sharing App for Universiti Malaya Community.**

Collaborator signature

Collaborator Name:
Position:
Collaborator Address:
Jabatan Hal Ehwal Pelajar (*Student Affairs Department*)
Aras 2, Blok E, Kompleks Perdanasiswa
Universiti Malaya
50603 Kuala Lumpur.
Date: 11-5-2026

FAIRUS AINI PAIRAN
Ketua Pusat Perkhidmatan Pelajar
Jabatan Hal Ehwal Pelajar
Universiti Malaya

Data Gathering Method (Survey)

YUM Hub

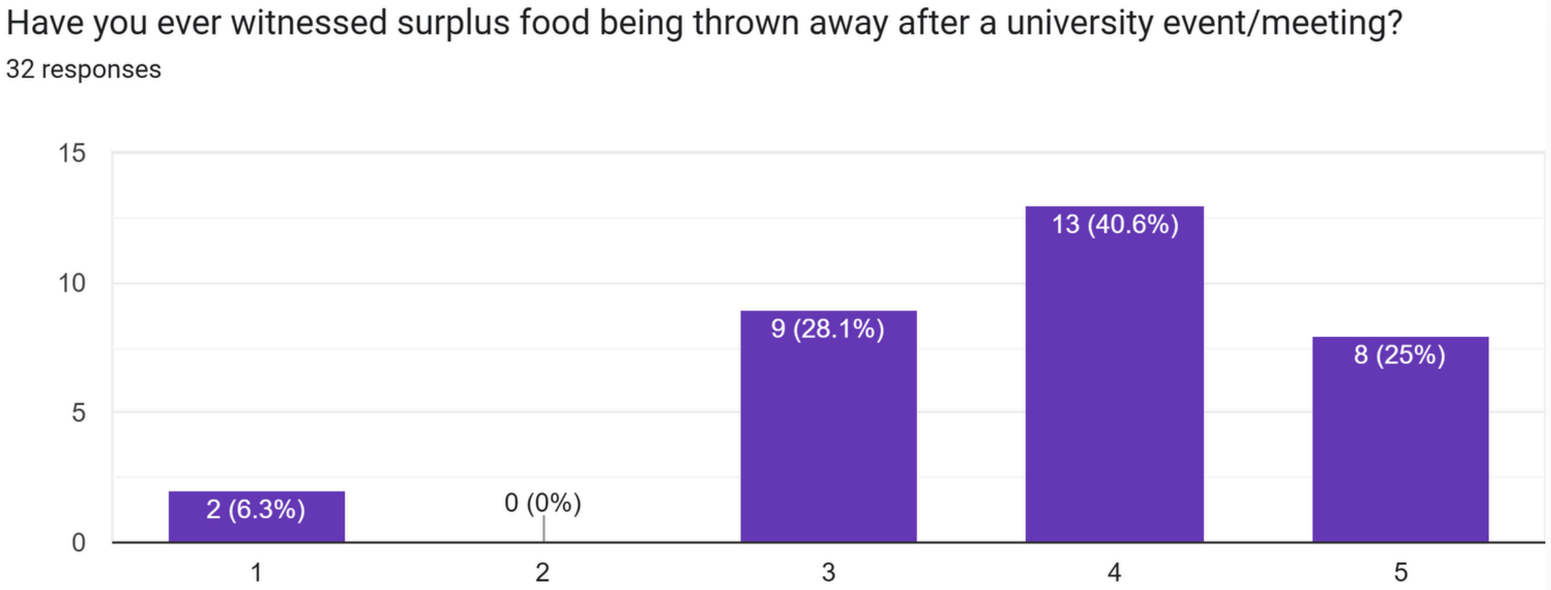
Survey Title : Student Food Insecurity & Awareness Survey

Food Waste & Accessibility Experience

Highest: Witnessing food waste after campus events (Level 4–5: 65.6%)

Lowest: Rare or no exposure to food waste (minor respondents))

Inference:
Food wastage is a frequent and visible issue on campus, indicating strong potential for a redistribution system to reduce surplus food disposal.



Data Gathering Method (Survey)

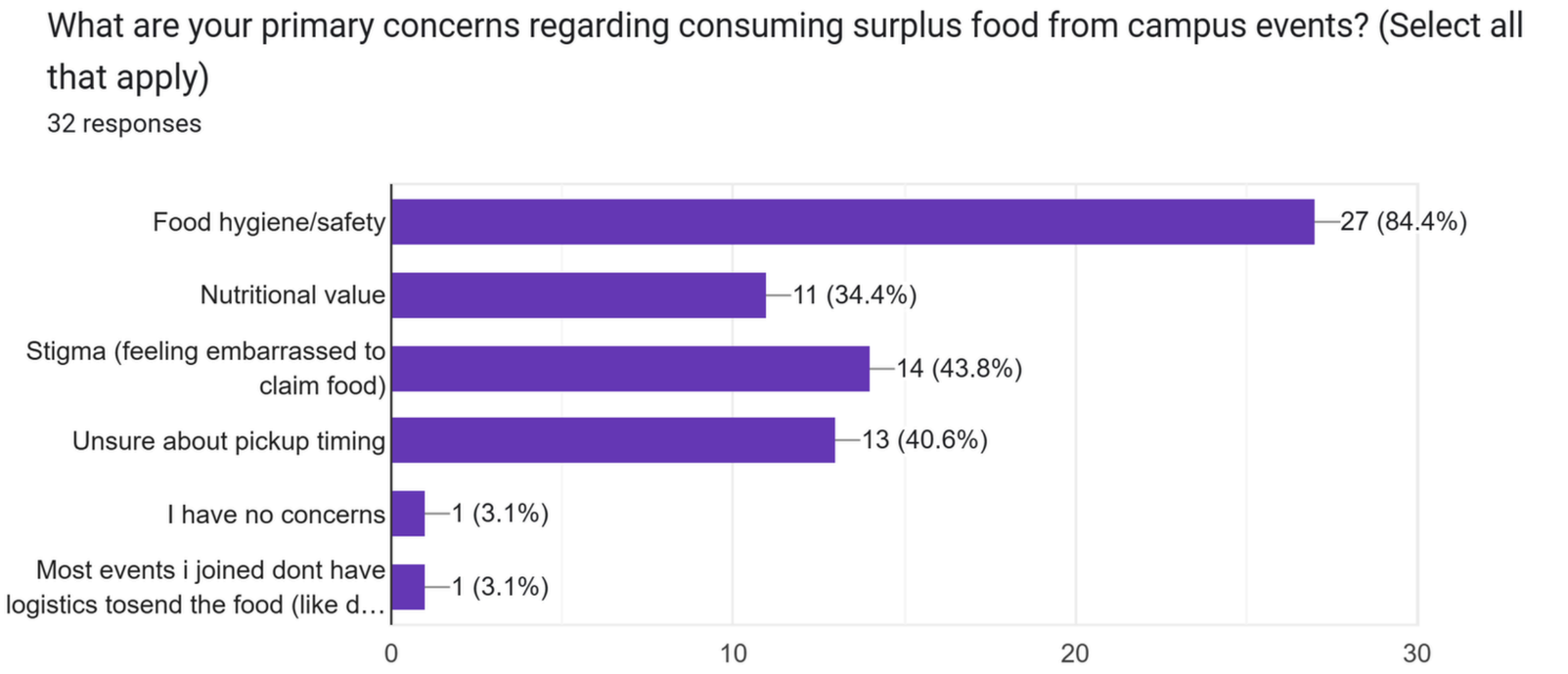
Survey Title : Student Food Insecurity & Awareness Survey

User Concerns & Essential Features

Highest: Food hygiene and safety concerns (84.4%)

Lowest: No concerns (6.2%)

Inference:
Safety is the dominant concern, requiring strict food verification, transparency, and trust-building features to ensure user adoption.



Data Gathering Method (Survey)

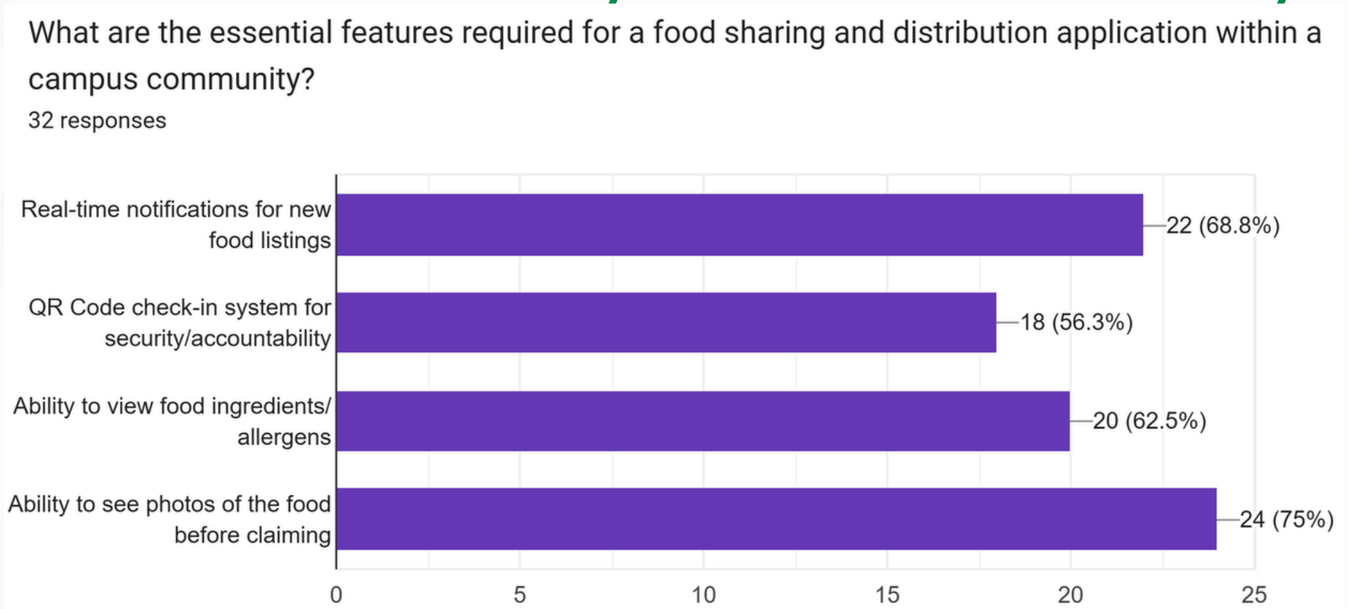
Survey Title : Student Food Insecurity & Awareness Survey

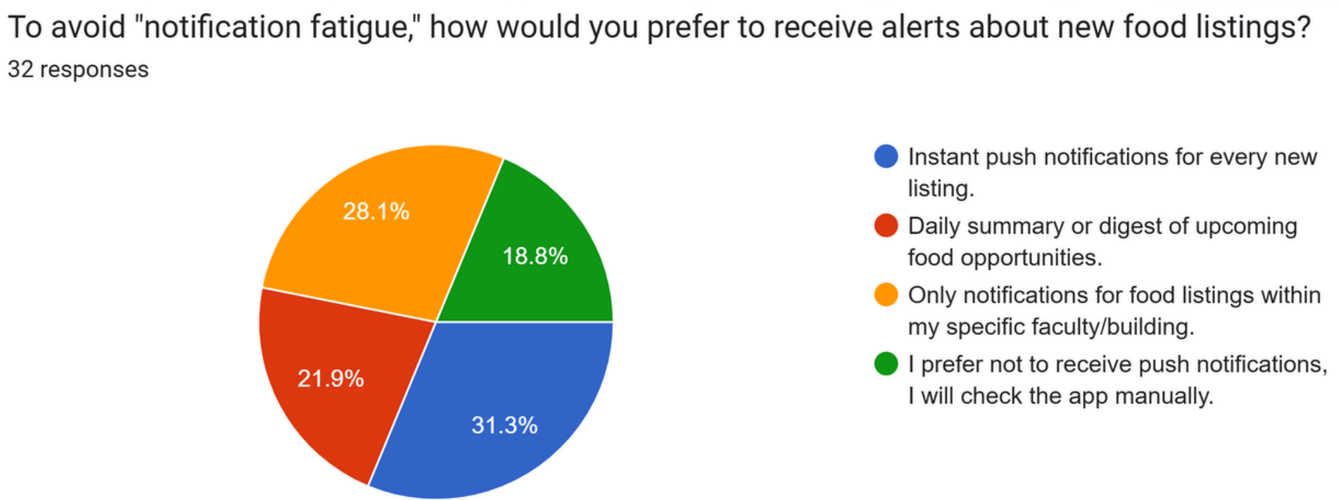
System Design & Feature Preferences

Highest: Viewing food photos before claiming (75%)

Lowest: Manual checking of app for updates (least preferred)

Inference:
Users prioritize visual transparency and real-time information, with QR verification and notifications also strongly preferred, highlighting demand for a fast, trust-based digital system.





Functional Requirements (Subtitle 2)

Module	ID	Functional Requirement	Use Case	Priority
Dashboard Module	FR-18	The system shall provide a dashboard that displays key summary information, including: • Total number of food listings • Total number of pax distributed • Total amount of food saved • Number of active users • Number of active locations • Number of today's active listings	UC-3 View Dashboard Summary	H
Food Listing Monitoring Module	FR-19	The system shall allow users to monitor real-time food sharing records, including: • Event name • Food type • Number of pax in total • Status of food • Time of distribution • Location of distribution	UC-18 Monitor Food Listings	H

Functional Requirements (Subtitle 2)

Module	ID	Functional Requirement	Use Case	Priority
Food Listing Monitoring Module	FR-20	The system shall allow admins to filter food listings by status.	UC-18 Monitor Food Listings	L
	FR-21	The system shall allow admins to remove or deactivate inappropriate food listings.	UC-19 Manage Food Listings	M
User Activity Monitoring Module	FR-22	The system shall allow admins to view ranking of active users based on participation level.	UC-20 Monitor User Activity	H
	FR-23	The system shall allow admins to view providers sharing records.	UC-20 Monitor User Activity	H

Functional Requirements (Subtitle 2)

Module	ID	Functional Requirement	Use Case	Priority
User Activity Monitoring Module	FR-24	The system shall allow admins to view users claiming records.	UC-20 Monitor User Activity	H
	FR-25	The system shall allow admins to restrict suspicious users.	UC-20 Monitor User Activity	M
Location Activity Monitoring Module	FR-26	The system shall allow admins to monitor most active food distribution locations.	UC-21 Monitor Location Activity	H
	FR-27	The system shall allow admins to monitor participation rates on each location.	UC-21 Monitor Location Activity	M

Functional Requirements (Subtitle 2)

Module	ID	Functional Requirement	Use Case	Priority
Reporting Module	FR-28	The system shall allow admins to generate reports.	UC-22 Generate Report	H
	FR-29	The system shall allow admins to export reports in PDF and CSV format.	UC-22 Generate Report	H

Requirements (Non-Functional Requirements)

ID	Quality Requirement	Quality Attributes
NFR-01	The system should encrypt the user's password before it is saved in the database.The system functionalities should only be accessible to authorized users based on their roles.Sensitive data, such as user profiles and personal information, should only be accessible to authorized users.	Security
NFR-02	The system should only allow specific users to modify certain data based on their roles.	Security
NFR-03	The system must be capable of supporting at least 500 concurrent users during peak periods (e.g., after large university events or lunch hours).	Scalability
NFR-04	The platform shall maintain an uptime of 99.5% daily to ensure food can be claimed at any time.	Availability

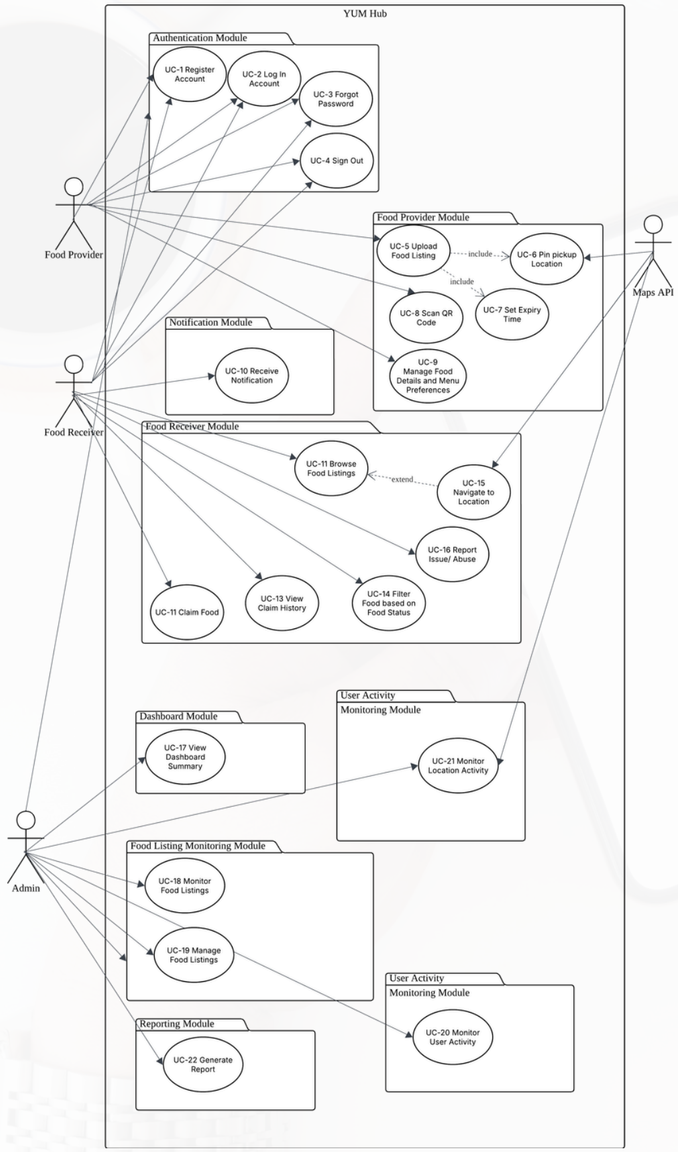
Requirements (Non-Functional Requirements)

ID	Quality Requirement	Quality Attributes
NFR-05	A first-time user shall be able to complete a "Food Claim" transaction in under 60 seconds without prior training or a manual.	Usability
NFR-06	The system should load dashboard data, charts and tables within 2-3 seconds under normal usage conditions.	Performance
NFR-07	The system shall ensure that all data is stored and processed without loss or corruption.	Security
NFR-08	The system shall be responsive and accessible across different screen sizes.	Usability

Requirements (Non-Functional Requirements)

ID	Constraint
NFR-01	The system is limited to covering only the Universiti Malaya main campus, Kuala Lumpur.
NFR-02	The system must exclusively use Universiti Malaya's email domain (@um.edu.my / @siswa.um.edu.my) for identity verification to ensure only campus members participate.
NFR-03	The application requires an active internet connection (WiFi/4G/5G) to sync inventory in real-time; no offline claiming mode is permitted to avoid over-pax distribution.

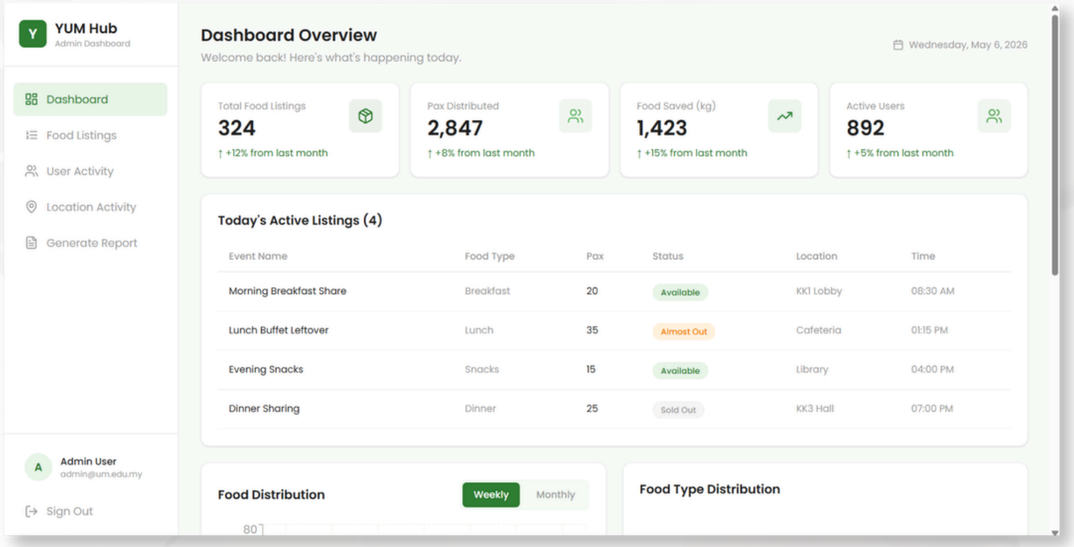
System Analysis & Design

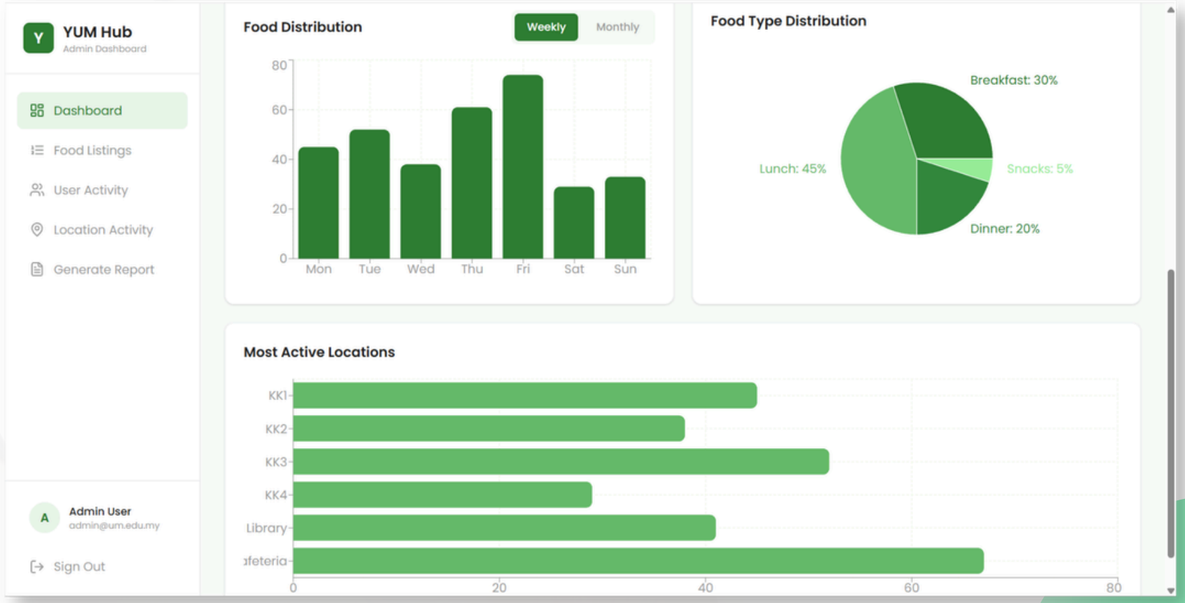


Use Case Diagram

System Analysis & Design

UI Mockup (Figma)

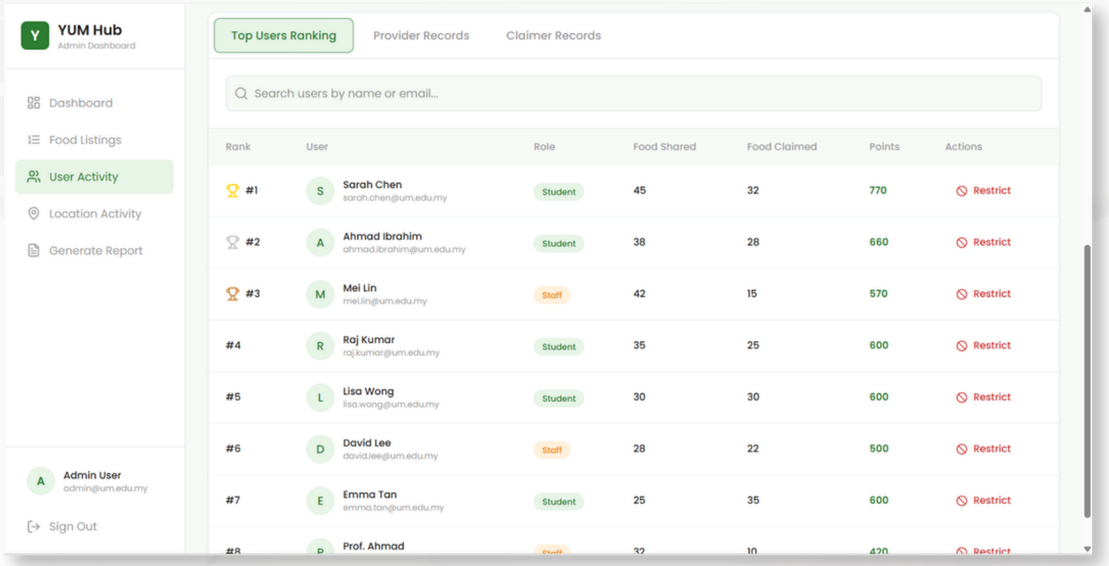


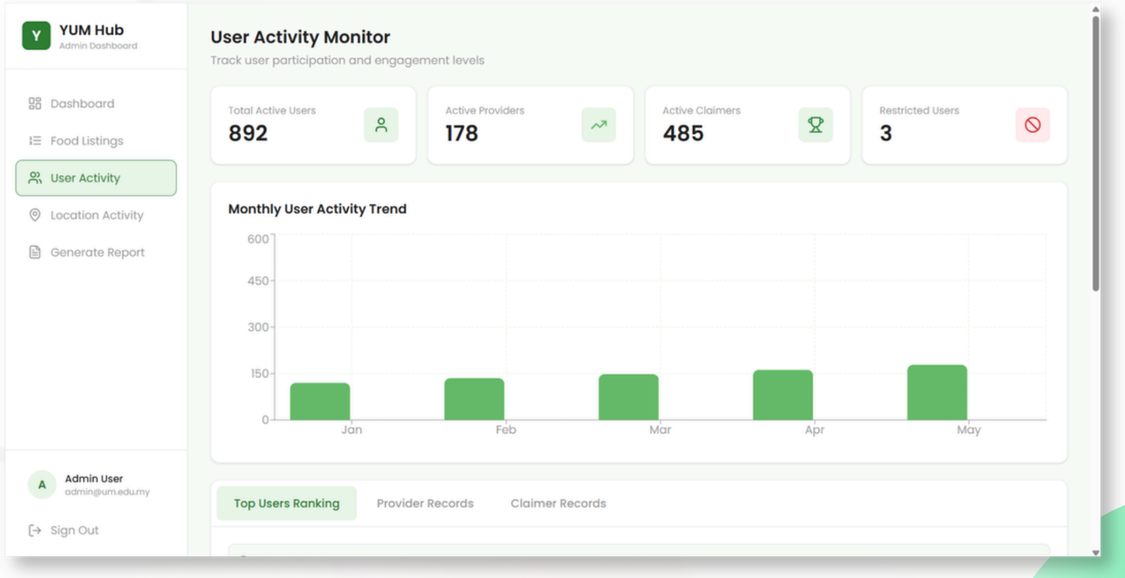


MAIN DASHBOARD

System Analysis & Design

UI Mockup (Figma)

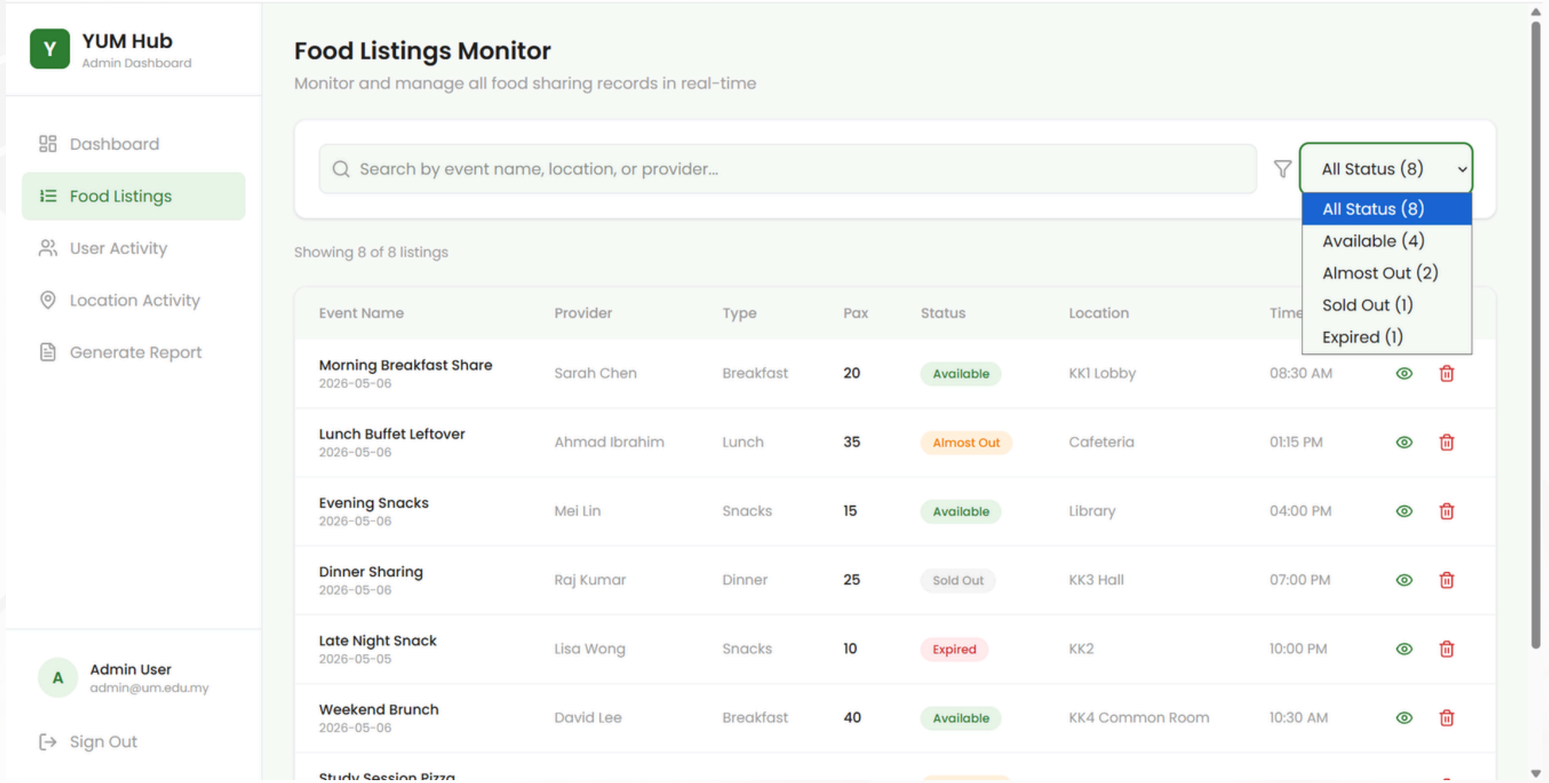




USER ACTIVITY

System Analysis & Design

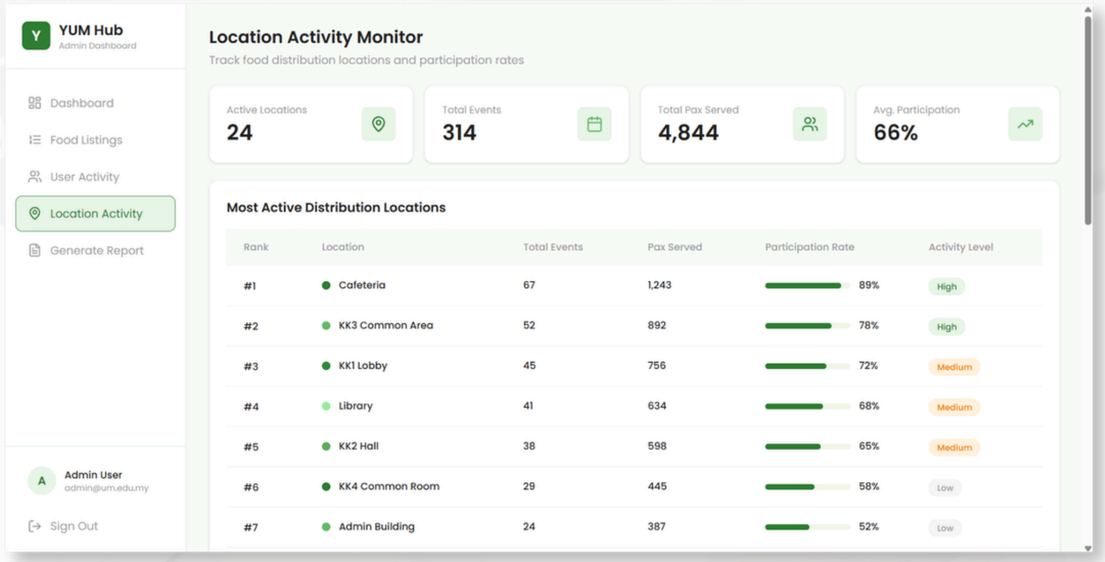
UI Mockup (Figma)



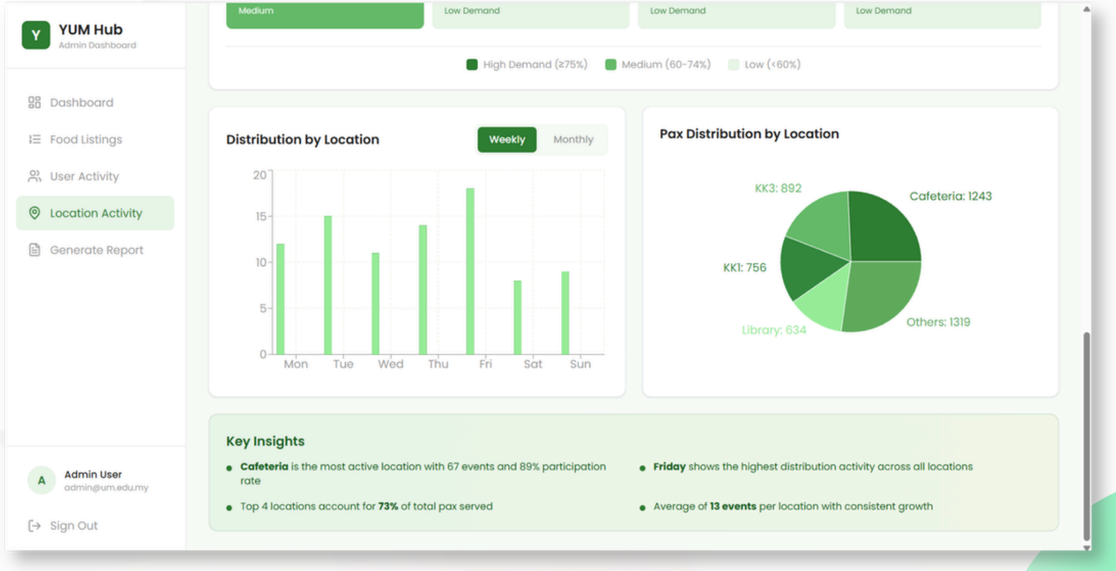
FOOD LISTINGS

System Analysis & Design

UI Mockup (Figma)

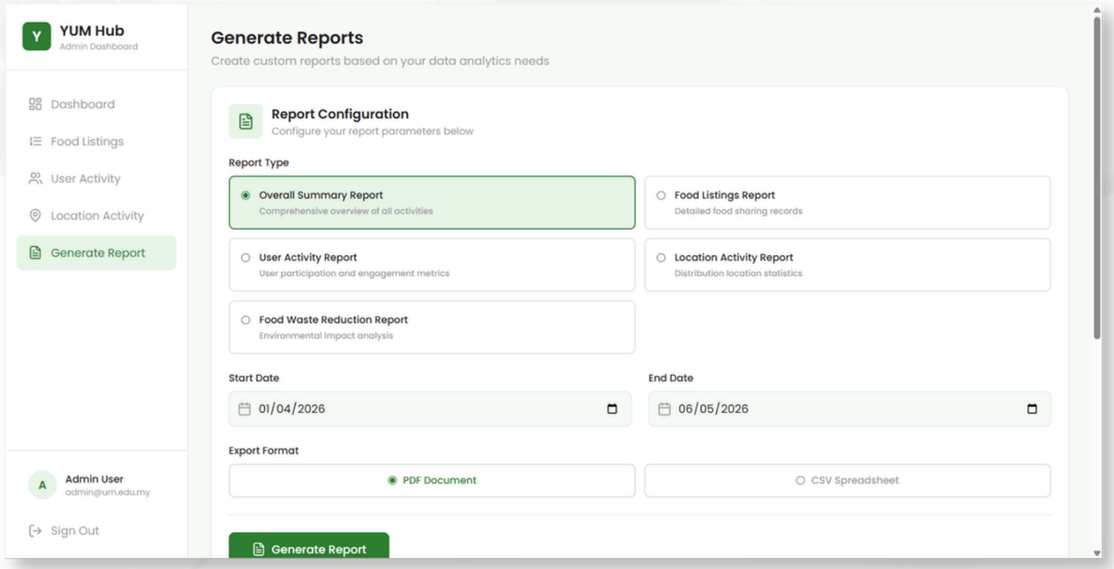


LOCATION ACTIVITY

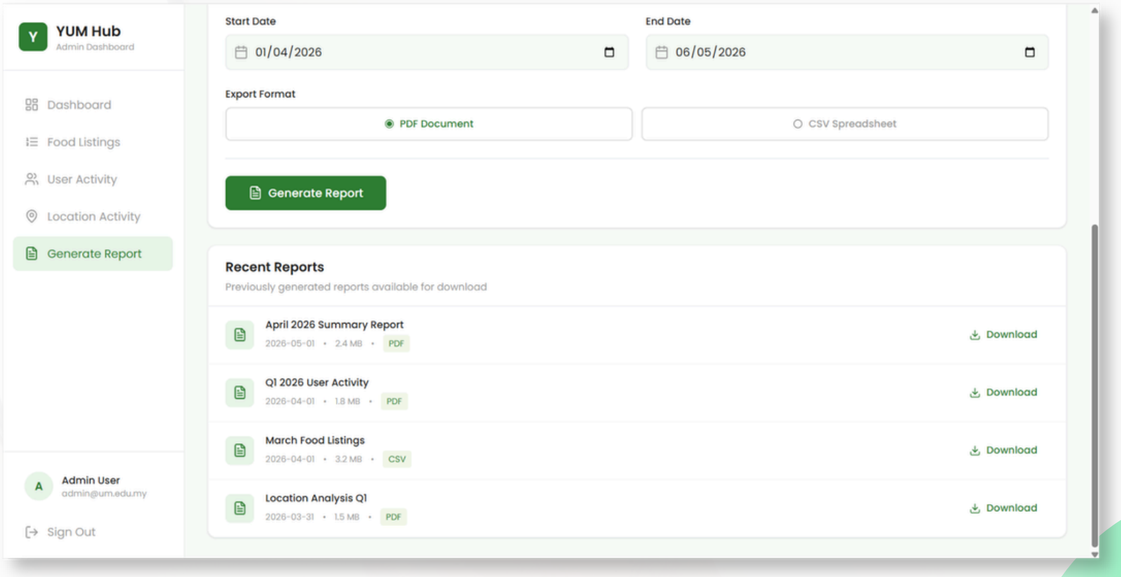


System Analysis & Design

UI Mockup (Figma)



GENERATE REPORT



Technical Implementations

Website

UI Layer

Next.js

TaildWind.CSS

Backend Integration Layer

Supabase PostgreSQL

Supabase Auth

Supabase Client API

Data Visualization

Chart.js / Rechart

Google Maps API

Reporting Implementation

jsPDF / CSV export

Thank You